

RUCHIR CHHEDA

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EDUCATION

Indian Institute of Technology, Bombay

2018 – 2023

- **B.Tech** in Electrical Engineering, **M.Tech** in Communication and Signal Processing (CPI: **9.09/10**)
- **Minors**: Computer Science, AI & Data Science

WORK EXPERIENCE

Adobe Systems | Machine Learning Engineer 2

July 2023 - Present

- Built **LangGraph-based AI agents** for multimodal asset analysis and pipeline orchestration
- Co-built an **image deduplication system** using structure analysis, color profiling, OCR, and language detection
- Engineered CLIP and LLM based solutions for image analysis including **shot, lighting and style detection**
- Collaboratively designed and developed pipelines for **video feature extraction** and **video similarity detection**
- Analysed **image** and **audio** capabilities using pre-annotated and vendor annotated datasets and worked on improvements
- Optimized ML model deployments and infrastructure for improved performance and reduced operational overhead

Adobe Systems | Machine Learning Intern

Summer 2022

- Unified multisource emotion datasets (**WEBEemo, Unsplash, Adobe Stock**) to enable CLIP-based sentiment analysis.
- Enhanced model performance through **prompt engineering, fine-tuning CLIP weights**, and adding **NN layers**
- Implemented dataset **outlier detection using K-means clustering** on CLIP embeddings and addressed dataset bias

Adobe Systems | Machine Learning Intern

Summer 2021

- Developed hierarchical time series forecasting models for Search Engine Marketing campaigns to predict **clicks, cost, and CPC** across **portfolio, campaign, and advertisement** levels
- Tested multiple forecasting models like **SARIMA, XGBoost, Prophet, and Uber Orbit** for ad performance prediction
- Implemented day-of-week bias correction using exponential decay and weighted averaging techniques

PROJECTS

Matsya, Autonomous Underwater Vehicle (AUV) | Tech Team

2018 - 2023

- **RoboSub, AUVSI & US Office of Naval Research** - Guide: Prof. Leena Vachhani, Prof. Hemendra Arya
AUV-IITB student team working on design and development of AUV for International AUVSI RoboSub competition
- **Achievements**: Joint Winners at SAVe 2019 (IIT Madras), Semi-finalists at RoboSub 2019, Young Researcher Prize IEEE OES 2021 (Japan), Finalist RoboSub 2022 (Maryland)
- **Research Head (2021-2023)**: Working on diverse multi-bot algorithms and inter vehicle communication simulations
- **Software Sub-Div Head (2020-2021)**: Led 8-member team, working on drag modelling and 3D image reconstruction
- **Software Developer (2019-2020)**: Developed state machine based AI algorithm and BFS Path Planning algorithm
- **Software Trainee (2018-19)**: Implemented user configurable timeouts, YAML checker, and system flow checkers

Unsupervised Prompt Learning | Master's Thesis

Jan 2023 - June 2023

- Explored **CLIP-based zero-shot classification** for satellite image datasets in unsupervised settings
- Improved pseudo-labelling and prompt optimization using **CoOp, CoCoOp, and AppleNet**
- Enhanced performance and domain generalization through **iterative pseudo-labeling** and **feature enhancement**

Semi Blind Image Super Resolution | Master's Thesis

Aug 2022 - Dec 2022

- Investigated **dictionary learning approaches** for reconstructing high-resolution images from low-resolution inputs
- Integrated **blurring kernel estimation** and **CNN-based enhancements** to improve reconstruction quality
- Demonstrated performance comparable to deep learning methods while requiring fewer training images

Health Screen Segmentation - Cloud Physician | Team Lead | Inter IIT

Gold

- Developed a pipeline for vital signs extraction using **Segformer** for screen segmentation and **YOLOv8** for detection
- Implemented OCR system using **Parseq** model with CLAHE, thresholding, erosion, and dilation preprocessing
- Engineered graph digitization algorithm using **contour detection**, **median filtering**, and **Otsu binarization**
- Enhanced YOLOv8 performance by 0.2% mAP-50 using CRAFT-based text score probability maps

Object Volume Estimation - Grow Simplee | Inter IIT

Silver

- Developed a pipeline using **OAK-D camera** to generate high-resolution depth maps for volume estimation
- Implemented **Monocular Depth Estimation (MDE)** using U-net architecture trained on RGB-Depth image pairs, converting depth maps to point clouds and calculating volume using **Qhull algorithm**

Age and Gender Detection - Bosch | Team Lead | Inter IIT

- Built a pipeline to detect faces in CCTV images using **RetinaFace** and extracted face crops for analysis.
- Enhanced face crops with super-resolution models (**SPARNet**, **FSRNet**, **GFPGAN**) and applied **ArcFace**, **Deepface** for age and gender prediction.
- **YOLO-based person tracking** implemented to ensure consistent identity assignment across video frames.

Multi-modal Facial Reconstruction System | IIT Bombay

- Built a multi-modal system using **GANs** to reconstruct realistic facial images from sketches and voice inputs
- Processed **VoxCeleb** and **VGGFace** data with face alignment and log-mel spectrogram extraction
- Evaluated effects of voice and sketch quality on reconstruction performance, identifying limitations and improvements

Neural Machine Translation (NMT) Project | IIT Bombay

- Implemented and benchmarked **RNN**, **GRU**, **CNN**, **Transformer**, and **Reformer** models in PyTorch for English-German/French translation
- Conducted experiments on **Multi30k** and **WMT14** datasets, analyzing model robustness using **BLEU scores**
- Derived efficiency-accuracy tradeoffs, establishing **Transformers** as optimal for resource-constrained NMT

OTHER WORK EXPERIENCE

SNL Innovations Pvt. Ltd. | Software Intern

Summer 2020

- Innofarms, SNL Innovations Pvt. Ltd. is a young startup working in the field of decentralized food processing to reduce wastage, increase farmers income and promote micro-enterprises
- Designed a **tri-party blockchain system** enabling transparent interaction between farmers, middlemen, and processors
- Built an **Ethereum-based B2B trading platform** with **smart contracts** and **IPFS**-backed document storage
- Developed backend services with **Flask** and a web interface using **web3.js**, integrating **Metamask wallet** support

DOCKABL | Software Intern

Summer 2020

- Clink is a platform that helps organizations to share active and valuable feedback with their employees
- Integrating **Clink** with **team-interaction platforms** like Slack in order to use the **Clink functionalities** on those platforms and **sending valuable actions** back to Clink for record

ACHIEVEMENTS

- Secured **3rd** position and rewarded with **\$400 cash prize** in **Ener'gentic Hackathon 2025**. Developed multi-agent assistant platform for sustainable energy onboarding with intelligent user guidance systems
- **Winner, Pan-IIT Blockchain Hackathon**: Developed cross-chain NFT bridge platform for Solana, Avalanche, and Ethereum networks. Built Flask web application on Heroku with ERC721 token bridging and multi-wallet connectivity
- Secured **2nd** position and rewarded with **INR 50k** in **Dremio Business Insight** competition on a **Big Data** problem
- Awarded change of branch to **Electrical Engineering** for academic excellence
- **JEE Advanced 2018**: All India Rank 1019 out of 1.64 Lakh candidates
- **JEE Mains 2018**: 99.90 Percentile out of 1.1 Million candidates
- **Intermediate/+2** (93.80%) & **Matriculation** (96.00%) from Delhi Public School, Navi Mumbai
- Recipient of **KVPY Fellowship 2017-18** by Department of Science and Technology, Government of India
- **National Top 1%** in NSE Physics & Chemistry (2018); Cleared NSE Astronomy (2017,18) & RMO(2017)